

Shubham Kumar

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Professional Summary

M.Tech candidate in Artificial Intelligence Machine Learning at Birla Institute of Technology, Mesra with hands-on experience in NLP, machine learning, and data science. Proficient in building end-to-end ML pipelines using Python, Scikit-learn, NumPy, and Pandas, with practical experience in deep learning using PyTorch and LLM application development using LangChain. Strong foundation in data structures, algorithms, and core computer science concepts, with 150+ solved coding problems. Passionate about building intelligent, data-driven systems and seeking AI/ML or software internships to apply and expand technical expertise.

Education

M.Tech – Artificial Intelligence & Machine Learning	CGPA: 8.1 (Ongoing)
Birla Institute of Technology (BIT) Mesra	2025 – 2027
GATE 2025 Qualified — CSIT Stream	
B.Tech – Computer Science & Engineering	CGPA: 7.79
Sarala Birla University, Ranchi	2021 – 2024

Technical Skills

- **Languages:** Python, C
- **ML / DL Libraries:** Scikit-learn, NumPy, Pandas, PyTorch (basics), NLTK
- **NLP & LLM:** Transformers (BERT), LangChain, Retrieval-Augmented Generation (RAG), FAISS, Word2Vec, TF-IDF
- **Core CS:** Data Structures & Algorithms, DBMS, Operating Systems, OOP
- **Tools & Platforms:** Git, GitHub, Streamlit, Jupyter Notebook, VS Code, Google Colab
- **Domains:** Machine Learning, NLP, Evolutionary Computing, Data Analysis, Clustering, Optimization

Projects & Academic Work

Word Sense Disambiguation using Lesk Algorithm Python, NLTK, WordNet

- Designed an NLP pipeline to resolve lexical ambiguity in text by leveraging the simplified Lesk algorithm with WordNet lexical databases.
- Implemented text preprocessing (tokenization, stop-word removal, lemmatization) to improve contextual overlap accuracy by 18%.
- Evaluated performance on SemEval benchmark datasets; outcome demonstrates applicability in chatbot and IR systems.

LULC Analysis using K-Means Clustering (Remote Sensing) Python, Scikit-learn, Rasterio, Sentinel-2 Imagery

- Built an unsupervised land-use/land-cover classification system processing multispectral Sentinel-2 satellite imagery using K-Means clustering.
- Applied spectral band engineering and image normalization to distinguish vegetation, water bodies, urban zones, and barren land.
- Visualized temporal LULC changes across multiple years to support environmental monitoring use-cases.

Sentiment Analysis on Product Reviews Python, NLTK, Scikit-learn, TF-IDF

- Developed a binary sentiment classifier (positive/negative) trained on e-commerce review datasets using TF-IDF features and Logistic Regression.
- Achieved 88%+ accuracy with cross-validated evaluation; explored Naive Bayes and SVM baselines for comparison.

DocChat – Retrieval-Augmented Generation (RAG) System for PDF Q&A Python, LangChain, FAISS, HuggingFace, Streamlit, Gemini

- Built an end-to-end modular RAG pipeline (Document Loader → Chunking → Embedding → Vector Store → Retriever → LLM) enabling natural-language question answering over PDF documents.
- Implemented configurable recursive text chunking and semantic embeddings (sentence-transformers/all-MiniLM-L6-v2) indexed with FAISS for top-k similarity retrieval.
- Integrated Google Gemini 2.5 Flash via LangChain with a context-grounded prompt template to minimize hallucination, and built an interactive Streamlit interface with adjustable chunking parameters, pipeline-status tracking, and retrieved-context visualization.

Achievements & Co-Curriculars

- Cleared GATE 2025 in AI/ML stream, demonstrating mastery of core CS and ML theory.
- Actively solving Striver A2Z DSA Sheet — 150+ problems across arrays, trees, graphs, and DP.
- Completed coursework in Evolutionary Computing covering GA, NSGA-II, MCTS, and multi-objective optimization.
- Contributed to a group research project implementing Niching GA, Multi-Objective GA, and Competitive Coevolution for job-scheduling optimization with unified visualizations.